

Problem sheet 2

1. A binomial random variable X has index $n = 10$ and parameter $p = 0.4$. Calculate,

a. $P(X = 0)$

b. $P(X = 3)$

c. $P(X < 3)$

$$a. P(X=0) = \frac{10!}{0! 10!} (0.4)^0 (1-0.4)^{10} = 0.6^{10} = 6.047 \times 10^{-3}$$

$$b. P(X=3) = \frac{10!}{3! 7!} (0.4)^3 (1-0.4)^7 = 0.2150$$

$$c. P(X < 3) = P(X=0) + P(X=1) + P(X=2)$$

$$= \frac{10!}{0! 10!} (0.4)^0 (1-0.4)^{10}$$

$$+ \frac{10!}{1! 9!} (0.4)^1 (1-0.4)^9$$

$$+ \frac{10!}{2! 8!} (0.4)^2 (1-0.4)^8$$

$$= 6.047 \times 10^{-3} + 4.031 \times 10^{-3} + 1.344 \times 10^{-3}$$

$$= 0.01142$$

2. Independent missiles are aimed at a target until two consecutive missiles hit it. Each missile hits the target with probability 0.7. Let X be the number of missiles sent. Evaluate $P(X = r)$ for $r = 2, 3$ and 4.

$$P(\text{hit}) = 0.7$$

$$P(X=2) = \frac{2!}{2! \cdot 0!} (0.7)^2 (1-0.7)^0 = 0.49$$

$$\begin{aligned} P(X=3) &= P(X=2 | \overline{\text{hit}}) P(\overline{\text{hit}}) = P(X=2) P(\overline{\text{hit}}) \\ &= 0.49 \cdot (1-0.7) = 0.147 \end{aligned}$$

$$\begin{aligned} P(X=4) &= P(X=3 | \overline{\text{hit}}) P(\overline{\text{hit}}) = P(X=3) P(\overline{\text{hit}}) \\ &= 0.147 \cdot (1-0.7) \stackrel{Y}{=} \frac{0.147}{0.0147} \\ &\quad + 0.7 \times 0.3 \times 0.7 \times 0.7 \end{aligned}$$

3. A bag contains 3 red counters and 5 white ones. A sample of four counters is drawn from the bag at random and the number, X , of red counters in the sample is counted. How many possible samples of 4 counters are there? For $r = 0, 1, 2$ and 3, calculate how many of these samples contain r red counters and $4 - r$ white counters. Hence find the probability distribution for X .

For red, P_r and for white, P_w

$$\therefore P_r(X=0) = P_w(X=4) = \frac{5}{8} \cdot \frac{4}{7} \cdot \frac{3}{6} \cdot \frac{2}{5} = \frac{1}{14} = \frac{\binom{3}{0}\binom{5}{4}}{70}$$

$$P_r(X=1) = P_w(X=3) = 4 \left(\frac{5 \cdot 4 \cdot 3 \cdot 2}{8 \cdot 7 \cdot 6 \cdot 5} \right) = \frac{3}{7} = \frac{\binom{3}{1}\binom{5}{3}}{70}$$

$$P_r(X=2) = P_w(X=2) = 6 \left(\frac{5 \cdot 4 \cdot 3 \cdot 2}{8 \cdot 7 \cdot 6 \cdot 5} \right) = \frac{3}{7} = \frac{\binom{3}{2}\binom{5}{2}}{70}$$

$$P_r(X=3) = P_w(X=1) = 4 \left(\frac{5 \cdot 3 \cdot 2 \cdot 1}{8 \cdot 7 \cdot 6 \cdot 5} \right) = \frac{1}{14} = \frac{\binom{3}{3}\binom{5}{1}}{70}$$

Pascal triangle, $\binom{4}{0}, \binom{4}{1}, \binom{4}{2}, \binom{4}{3}$ and $\binom{4}{4}$ not included

$$* \binom{8}{4} = 70 \text{ possible samples}$$

4. The number of α -particles emitted in a one second interval by 1 gram of a radioactive substance has a Poisson distribution with rate 3.2 particles per second. What is the probability that no more than 2 particles will be emitted in a one second interval?

$$\lambda = 3.2 \text{ s}^{-1}$$

$$t = 1 \text{ s}$$

$$\therefore \mu = \lambda t = 3.2$$

$$\therefore P(X \leq 2)$$

$$= P(X=0) + P(X=1) + P(X=2)$$

$$= \frac{e^{-3.2} (3.2)^0}{0!} + \frac{e^{-3.2} (3.2)^1}{1!} + \frac{e^{-3.2} (3.2)^2}{2!}$$

$$= 0.3799$$

5. Suppose that an aircraft engine will fail, when in flight, with probability 0.001 and that different engines are independent. If an aircraft needs the majority of its engines operative to make a successful flight, and the probability of a successful flight for a 3-engined aircraft and a 5-engined aircraft.

$$p = 0.001$$

for 3-engined aircraft.

$$P(X=0) = \binom{3}{0} (0.001)^0 (0.999)^3 = 0.9970$$

$$P(X=1) = \binom{3}{1} (0.001)^1 (0.999)^2 = 2.994 \times 10^{-3}$$

$$\therefore P(X \leq 1) = P(X=0) + P(X=1) = 0.999994 \approx 1$$

for 5-engined aircraft.

$$P(X=0) = 0.9950$$

$$P(X=1) = 4.980 \times 10^{-3}$$

$$P(X=2) = 9.970 \times 10^{-6}$$

$$\therefore P(X \leq 2) = 0.99998997 \approx 1$$

6. The time, T months, required to build an aircraft component has an exponential distribution with parameter $\lambda = 0.5$. Calculate,
- The probability that it takes less than 1 month to build the component.
 - The probability that it takes longer than 3 months to build the component.
 - The probability that it takes longer than 3 months to build the component given that the work has already lasted 2 months.

$$\begin{aligned}
 \text{a. } P(T < 1) &= \int_0^1 0.5e^{-0.5t} dt \\
 &= [-e^{-0.5t}]_0^1 \\
 &= (1 - e^{-0.5}) = 0.3935
 \end{aligned}$$

$$\begin{aligned}
 \text{b. } P(T > 3) &= 1 - P(T \leq 3) = 1 - P(T < 3) \\
 &= 1 - (1 - e^{-1.5}) = 0.2231
 \end{aligned}$$

$$\begin{aligned}
 \text{c. } P(T > 3 \mid T \geq 2) &= P(T > \overset{1}{\cancel{3}}) = \overset{1-}{\underset{\wedge}{0.3935}} \\
 &= 0.6065
 \end{aligned}$$

7. The flying time, T , between two cities A and B , has approximately a normal distribution with $\mu = 4$ and $\sigma = 0.5$ hours. Find the probability that a flight between A and B ,
- takes longer than 5 hours.
 - takes between 3.5 and 5 hours.

$$\begin{aligned} \text{a. } P(X > 5) &= 1 - P(X < 5) \\ &= 1 - \Phi\left(X < \frac{5-4}{0.5}\right) \\ &= 1 - \Phi(X < 2) \\ &= 1 - 0.9772 \\ &= 0.0228 \end{aligned}$$

$$\begin{aligned} \text{b. } P(3.5 \leq X \leq 5) &= P(X < 5) - P(X < 3.5) \\ &= \Phi(X < 2) - \Phi(X < -1) \\ &= \Phi(X < 2) - [1 - \Phi(X < 1)] \\ &= 0.9772 - (1 - 0.8413) \\ &= 0.8185 \end{aligned}$$

8. The thickness of manufactured metal plates is approximately normally distributed with $\mu = 20$ mm and $\sigma = 0.04$ mm. How would you have to set the tolerance limits $20 - c$, and $20 + c$ such that at most 5% scrap is produced? Assume now that μ has changed to 20.10 mm. Find the percentage of metal plates that exceed your previously calculated tolerance limits.

For at most 5% scrap is produced,

$$P(20 - c \leq X \leq 20 + c) = 1 - 5\% = 95\%$$

$$\therefore P(X \leq 20 + c) - P(X \leq 20 - c)$$

$$= \Phi(X < 25c) - \Phi(X < -25c)$$

$$= \Phi(X < 25c) - [1 - \Phi(X < -25c)]$$

$$= 2\Phi(X < 25c) - 1 = 0.95$$

$$\therefore \Phi(X < 25c) = 0.975$$

$$\therefore 25c = 1.96 \quad \therefore c = 0.0784$$

for $\mu = 20.10$ mm,

$$P(20 - c \leq X \leq 20 + c)$$

$$= \Phi(-4.46 < X < -0.54) = \Phi(0.54 < X < 4.46)$$

$$= \Phi(X < 4.46) - \Phi(X < 0.54) = 1 - 0.7054 = 0.2946$$

$$\therefore 1 - 0.2946 = 70.54\%$$

9. By switching to polar coordinates show that

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x^2+y^2)} dx dy = 2\pi$$

Using this result, prove that if $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$ is the p.d.f of the standard normal distribution, then

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x^2+y^2)} dx dy$$

$$= \int_0^{2\pi} \int_0^{\infty} e^{-\frac{1}{2}r^2} r dr d\theta$$

$$\text{Let } u = \frac{1}{2}r^2 \quad \therefore \frac{du}{dr} = r$$

$$\therefore \int e^{-\frac{1}{2}r^2} r dr = \int e^{-u} du = [-e^{-u}] = [-e^{-\frac{1}{2}r^2}]$$

$$\therefore \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x^2+y^2)} dx dy$$

$$= \int_0^{2\pi} [-e^{-\frac{1}{2}r^2}]_0^{\infty} d\theta$$

$$= \int_0^{2\pi} 1 d\theta$$

$$= 2\pi$$

$$\text{Let } I = \int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx$$

$$\therefore I^2 = \int_{-\infty}^{\infty} f(x) dx \int_{-\infty}^{\infty} f(y) dy$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x^2+y^2)} dx dy$$

$$= \frac{1}{2\pi} \cdot 2\pi$$

$$= 1$$

10. Let $X \sim N(a, b)$. What is the distribution of the following:

- a. $Y = 1 + X$
- b. $Y = 2X$
- c. $Y = 100 + 4X$
- d. $Y = cX - 0.1$

a. $Y \sim N = (a+1, b)$

b. $Y = X + X$

$$\therefore Y \sim N = (a + a, \overset{2 \times b}{\cancel{b + b}}) = (2a, \overset{4}{\cancel{2}b})$$

c. $Y \sim N = (4a + 100, \overset{16}{\cancel{4}b})$

d. $Y \sim N = (ac - 0.1, bc^2)$